Subnetting class A Networks (Day-15)

You have been given the 10.0.0.0/8 network. You must create 2000 subnets which will distributed to various enterprises. What prefix length must you use? How many n addr. (usable addr) will be in each subnet?

00001010 . 00000000 . 00000000 . 00000000
10 . 0 . 0 . 0

For 2000 subnets, no. of bits to burrow = 11.
$2^{11} = 2048$
So, our prefix length is /19
So, no. of host bits = 13
No. of hosts per subnet = 8190

FLSM (Fixed Length Subnet Mask)

- Till now, we have practiced subnetting using FLSM
- This means that all of the subnets use the same prefix length (i.e. subnetting a class C network into 4 subnets using /26).

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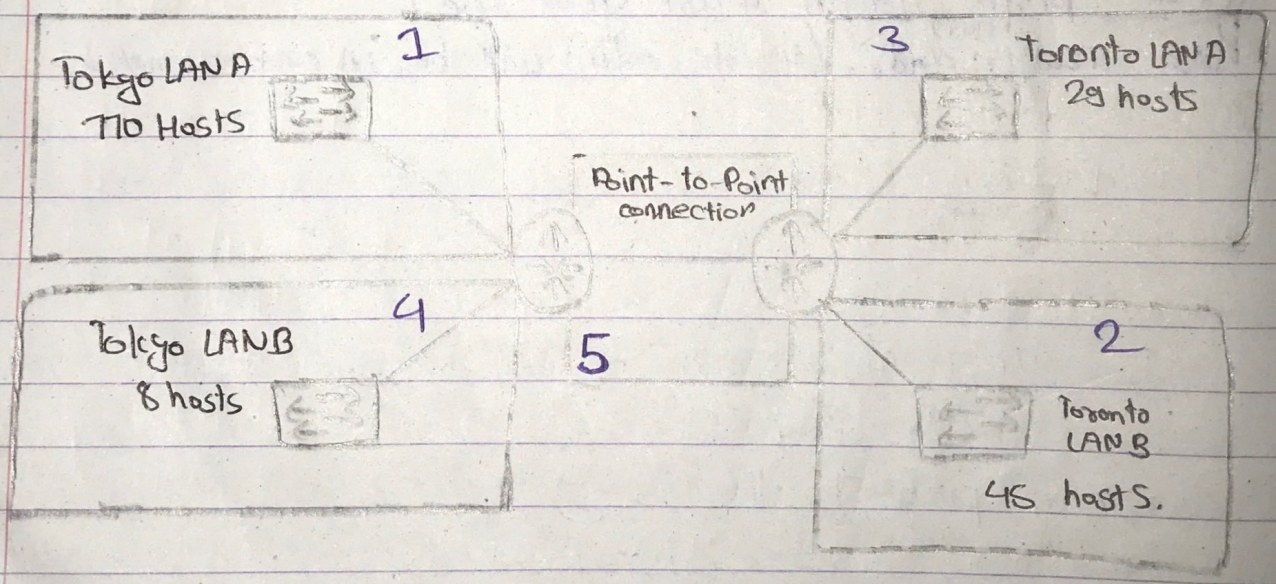
VLSM (Variable Length Subnet Mask)

- VLSM is the process of creating subnets of different sizes, to make ~~your~~ use of network addresses more efficient.

- VLSM is complicated than FLSM, but its easy if you follow the steps correctly.

Subnetting a network using VLSM.

192.168.1.0/24



We're assigned the 192.168.1.0/24 network, and must divide it into 5 subnets to provide IP address address for all hosts in the enterprise network.

If we were to try this with fixed-length subnet masks, we would need to borrow 3 bits to make enough subnets. That would leave us with 5 host bits.

5 host bits = 30 usable address, which ain't enough for Tokyo LAN & or Tokyo LAN Toronto LAN B.

If we use VLSM we can assign different subnet sizes to each LAN, which will allow us to ensure each LAN has enough addresses available.

Steps to Subnet:

i) Assign the largest subnet at the start of address space.
ii) Assign the second-largest subnet after it.
iii) Repeat the process untill all subnets have been assigned

Let's see our network for eg.

i) Tokyo LAN A then Torronto lan B, then Torronto lan A & finally Tokyo lan B

1a) Tokyo-LAN A

192.168.1.0 → 11000000.10101000.00000001.00000000

Prefix Length: /25
Network add: 192.168.1.0/25
Broadcast add: 192.168.1.127/25
First usable add: 192.168.1.1/25
Last usable add: 192.168.1.126/25
Total no. of usable @ host address: $2^7-2 = 126$

LAN A has 110 hosts $2^6 < 110 < 2^7$. So, we'll need 7 host bits, so that means we'll borrow 1 bit for network portion & our prefix length'll be /25

2b) (Torronto) Tokyo LAN B.

LAN A occupies half of address of our network. But using VLSM we'll assign other remaining addresses to smaller subnets.

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192.168.1.127 was broadcast for Tokyo A. So network for Torront B would be 192.168.1.128. We need 45 host

$32 < us < 64$, $2^5 < us < 2^6$, ie 6 hosts bits needed

So, prefix length is /26.

Net

Network add: 192.168.1.128 /26
Broadcast add: 192.168.1.191 /26
No. of usable hosts: 62
First usable add: 192.168.1.129 /26
Last usable add: 192.168.1.190 /26.

#3) Torronto LAN A.

As 192.168.1.191 was broadcast of subnet 2 [Torronto B). Network add of Torronto A = 192.168.1.192 . Also hosts = 29, So host bit = 5

Prefix length = /27
Network add: 192.168.1.192 /27
Broadcast add: 192.168.1.223 /27
First usable add: 192.168.1.193/27.
Last usable add: 192.168.1.222/27
Total no. of usable host = 30

#4) Tokyo LAN B.

Tokyo LAN B requires 8 host, so so we'll need 4 host bits not 3 coz, in 3 bits $8-2=6$ usable address.

Prefix length = /28
Network add = 192.168.1.224/28
Broadcast = 192.168.1.239/28

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First usable add: 192.168.1.225/28
Last usable add: 192.168.1.238/28
Total no of usable host address: 14.

#5 Point-Point Connection.

Network address = 192.168.1.240/30
Prefix length. /30
First usable add: 192.168.1.241/30
Last usable add: 192.168.1.242/30
Total no. of host add: 2
~~No of tick~~ Broadcast add: 2

For CCNA, if we've asked what prefix length to use for a subnet that requires two hosts (like ours), ~~yes but~~ don't say /31 say /30

Additional Practice

1. Subnettingquestions.com
2. subnetting.org
3. Subnettingpracctice.com